

S-X2 against conditional infit, revised

Detection and false flags at a common family-wise error rate

Magnus Johansson

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Table of contents

Purpose	1
Design	1
Results	2
Under the null	3
Detection of the misfitting items	4
False flags on fitting items	5
Recovering the right items	6
Detection by item location	7
At a matched null error rate	8
Paired comparison	8
Interpretation	8
Scope and limits	9
Appendix: result tables	9
Reference	15

Purpose

An earlier study (July 2026, since removed) compared the generalized S-X2 index with conditional infit and item-restscore and concluded that S-X2 is a large-sample complement to infit. A review of that study found seven problems. The two that matter most are that infit was run without multiplicity correction while S-X2 used BH, and that mirt's S-X2 p-values are liberal under a true Rasch model. `sx2_null_diagnostic.qmd` traced the second to a df one too small and found a smaller far-tail excess that remains after correcting it.

This study asks the original question again with every method held to a family-wise error rate: does S-X2 detect misfit that infit misses, or flag fewer fitting items when several items misfit? The design and the decision rule were written before the run, in `sx2-comparison-v2-design.md`. The simulation is `sx2-comparison-v2.R`. Item-restscore is left out, since its comparison with infit is covered by `restscore_infit_power.qmd` and `restscore_power_conditional.qmd`.

Design

- 20 dichotomous items, locations evenly spaced on $[-2, 2]$.
- Persons from a bivariate normal, both dimensions SD 1.5, correlation ρ . Fitting items respond to the first dimension, misfitting items to the second. Fresh data in every replication.
- $k = 1$ misfitting item (nearest 0 logits) or $k = 3$ (nearest 0, -1 and -2). $\rho = 0.15$ is strong multidimensionality, 0.50 and 0.75 are subtle.
- $n = 150, 500, 1000$ and 2000 .
- 500 replications in each null cell ($k = 0$), 200 in each misfit cell. The Monte Carlo SE of a 5 percent family-wise rate is 1.0 point.

Label	Method
infit_wy	conditional infit, RMitemInfitCutoff() B = 400, Westfall-Young, the package default
infit_bh	same bootstrap, BH on the marginal p
sx2_mirt	S-X2, chi-square on mirt's df, BH
sx2_df1	S-X2, chi-square on df + 1, BH
sx2_boot	S-X2, parametric bootstrap B = 400, Westfall-Young
rmsea_boot	RMSEA.S_X2, same bootstrap draws, Westfall-Young
rmsea_05	RMSEA.S_X2 above a fixed .05

```

long <- do.call(rbind, lapply(L, function(x) do.call(rbind, lapply(seq_along(x$res),
function(r) {
  y <- x$res[[r]]
  if (is.null(y$res)) return(NULL)
  cbind(k = x$cell$k, rho = x$cell$rho, n = x$cell$n, rep = r, y$res)
}))))

methods <- c("infit_wy", "infit_bh", "sx2_mirt", "sx2_df1", "sx2_boot",
             "rmsea_boot", "rmsea_05")

adj <- long |>
  group_by(k, rho, n, rep) |>
  mutate(infit_wy = padj_infit, infit_bh = p.adjust(p_infit, "BH"),
         sx2_mirt = p.adjust(p_mirt, "BH"), sx2_df1 = p.adjust(p_df1, "BH"),
         sx2_boot = padj_sx2boot, rmsea_boot = padj_rmseaboot) |>
  ungroup()

flags <- adj |>
  mutate(across(c(infit_wy, infit_bh, sx2_mirt, sx2_df1, sx2_boot, rmsea_boot),
                 \ (p) p < .05),
         rmsea_05 = rmsea > .05) |>
  pivot_longer(all_of(methods), names_to = "method", values_to = "flag") |>
  mutate(flag = coalesce(flag, FALSE),
         method = factor(method, levels = methods))

# kable with rho printed to two decimals, so digits = 1 does not round it
kb <- function(d, ...) {
  d <- mutate(d, across(any_of("rho"), \ (x) sprintf("%.2f", x)))
  kable(d, ...)
}

```

Every replication returned results for every method. No bootstrap draw was dropped. easyRasch2 1.3.1.9002, mirt 1.46.1, about 7.6 hours on 10 cores.

Results

Each figure shows one method per colour and point shape, with the same colour for a method in every figure. The values behind every figure are in the appendix.

```

pal <- c(infit_wy = "#2a78d6", sx2_df1 = "#eb6834", sx2_boot = "#1baf7a",
        sx2_mirt = "#eda100", rmsea_05 = "#e87ba4")
shp <- c(infit_wy = 16, sx2_df1 = 17, sx2_boot = 15, sx2_mirt = 18,
        rmsea_05 = 4)
lab <- c(infit_wy = "Infit, Westfall-Young",
        sx2_df1 = "S-X2, df + 1, BH",
        sx2_boot = "S-X2, bootstrap, Westfall-Young",
        sx2_mirt = "S-X2, mirt df, BH",
        rmsea_05 = "RMSEA > .05")

cell_lab <- function(d) d |>
  mutate(cell = factor(sprintf("k = %d, rho = %.2f", k, rho),
                           levels = c("k = 1, rho = 0.50", "k = 1, rho = 0.75",
                                       "k = 3, rho = 0.15", "k = 3, rho = 0.50",
                                       "k = 3, rho = 0.75")))

rate_plot <- function(d, ylab) {
  ggplot(d, aes(factor(n), v, colour = method, shape = method,
                    group = method)) +
    geom_line(linewidth = 0.6) +
    geom_point(size = 2.2) +
    scale_colour_manual(values = pal, labels = lab, name = NULL) +
    scale_shape_manual(values = shp, labels = lab, name = NULL) +
    scale_y_continuous(limits = c(0, 100)) +
    labs(x = "Sample size", y = ylab) +
    theme_minimal(base_size = 11) +
    theme(legend.position = "bottom", panel.grid.minor = element_blank()) +
    guides(colour = guide_legend(nrow = 2), shape = guide_legend(nrow = 2))
}

fig_methods <- c("infit_wy", "sx2_df1", "sx2_boot", "rmsea_05")

```

Under the null

```

null_fw <- flags |>
  filter(k == 0) |>
  group_by(method, n, rep) |>
  summarise(any = any(flag), .groups = "drop") |>
  group_by(method, n) |>
  summarise(v = 100 * mean(any), .groups = "drop")

mcse <- 100 * 2 * sqrt(.05 * .95 / 500)
null_fw |>
  filter(method %in% c("infit_wy", "sx2_mirt", "sx2_df1", "sx2_boot")) |>
  ggplot(aes(factor(n), v, colour = method, shape = method, group = method)) +
  annotate("rect", xmin = -Inf, xmax = Inf, ymin = 5 - mcse, ymax = 5 + mcse,
          fill = "grey85", alpha = 0.6) +
  geom_hline(yintercept = 5, linetype = "dashed", colour = "grey40") +
  geom_line(linewidth = 0.6) +
  geom_point(size = 2.2) +
  scale_colour_manual(values = pal, labels = lab, name = NULL) +
  scale_shape_manual(values = shp, labels = lab, name = NULL) +
  scale_y_continuous(limits = c(0, 14), breaks = seq(0, 14, 2)) +
  labs(x = "Sample size", y = "Datasets with any flag (%)") +

```

```
theme_minimal(base_size = 11) +
theme(legend.position = "bottom", panel.grid.minor = element_blank()) +
guides(colour = guide_legend(nrow = 2), shape = guide_legend(nrow = 2))
```

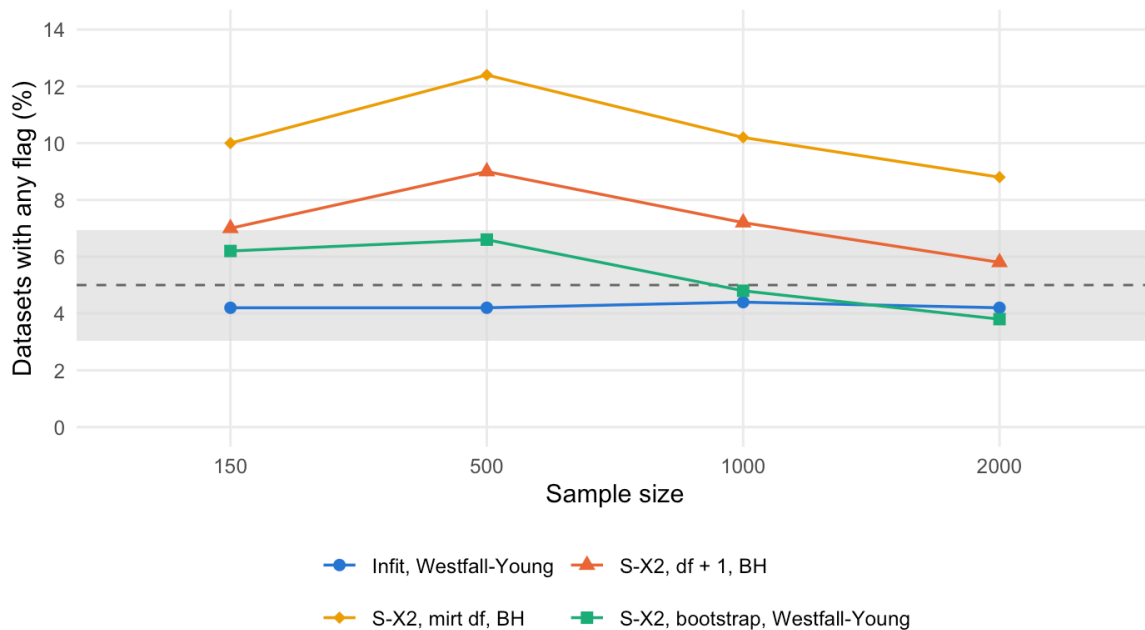


Figure 1: Share of datasets with at least one flagged item under a true Rasch model ($k = 0$, 500 datasets per n). The dashed line is the nominal 5 percent and the band covers two Monte Carlo SEs around it. RMSEA above .05 is left out, since it flags 99.2 percent of datasets at $n = 150$.

Infit with Westfall-Young holds 4.2 to 4.4 percent at every n . The S-X2 bootstrap gives 6.2 and 6.6 percent at $n = 150$ and 500 and 4.8 and 3.8 percent at $n = 1000$ and 2000. The first two are 1.2 and 1.6 Monte Carlo SEs above 5, so it is at most slightly liberal at small n . With the df corrected, asymptotic S-X2 gives 5.8 to 9.0 percent, in line with the diagnostic. mirt's default gives 8.8 to 12.4 percent, and RMSEA above .05 flags almost every dataset at $n = 150$ and none at $n = 2000$ (Table 1).

RMSEA with a bootstrap cutoff behaves like the S-X2 bootstrap in every cell of this study, as expected, since both are computed from the same draws and RMSEA is close to a monotone function of S-X2. It is left out of the figures. Infit with BH on the marginal bootstrap p is close to infit with Westfall-Young and is left out too. Both are in the appendix tables.

Detection of the misfitting items

```
flags |>
  filter(k > 0, misfit, method %in% fig_methods) |>
  group_by(k, rho, n, method) |>
  summarise(v = 100 * mean(flag), .groups = "drop") |>
  cell_lab() |>
  rate_plot("Detection (%)") +
  facet_wrap(~ cell, nrow = 2)
```

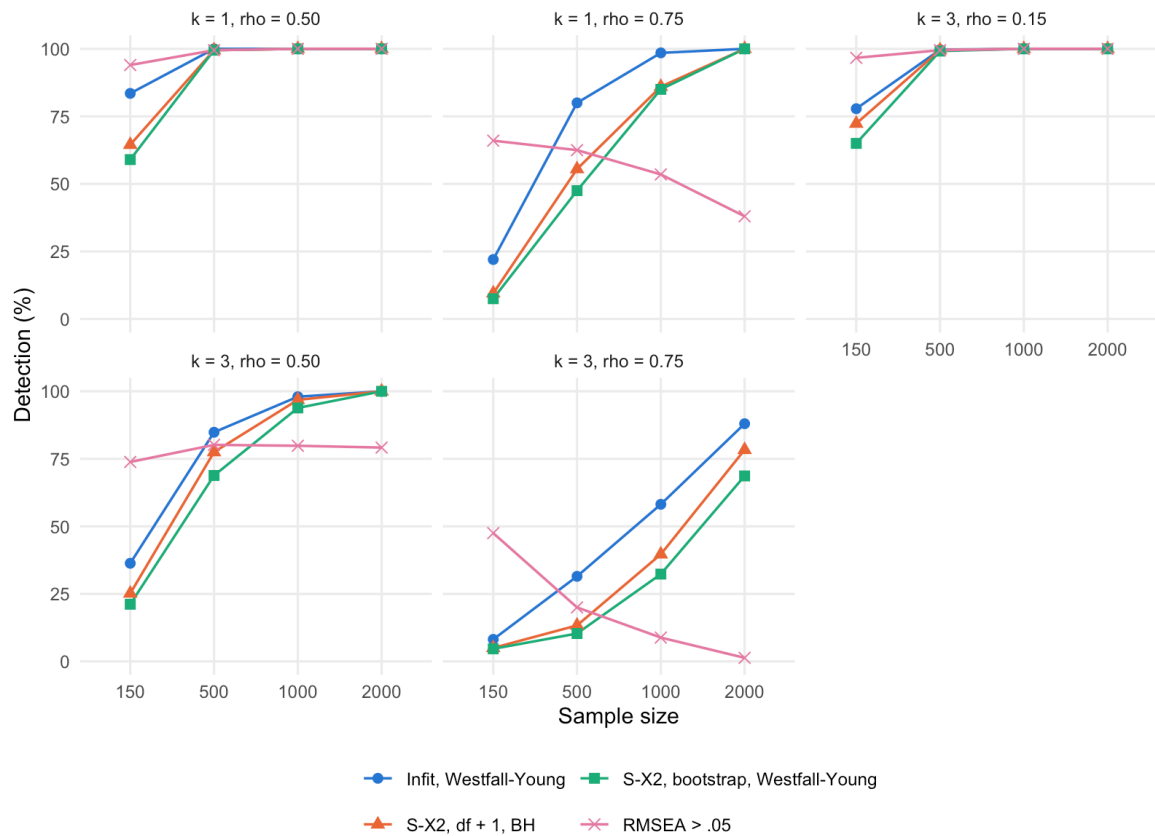


Figure 2: Detection of the misfitting items, averaged over the misfitting items in each dataset. k is the number of misfitting items and ρ the correlation between the two dimensions, lower meaning stronger misfit.

Infit with Westfall-Young detects at least as much as the S-X2 bootstrap in every cell, and clearly more wherever detection is not saturated. The gap is largest for subtle misfit. At $\rho = 0.75$ it is 80 against 47.5 percent with one misfitting item at $n = 500$, and 58 against 32 percent with three at $n = 1000$. The S-X2 bootstrap only catches up where both methods reach 100 percent (Table 2).

The fixed RMSEA threshold detects more at $n = 150$ because it flags almost everything there, and falls to 1.3 percent at $\rho = 0.75$ and $n = 2000$, as in the original study. A bootstrap cutoff for RMSEA recovers that, 68.7 percent in the same cell, which confirms the hypothesis in conclusion 5 of the original study. It does no better than the S-X2 bootstrap, though.

False flags on fitting items

```
flags |>
  filter(k > 0, !misfit, method %in% fig_methods) |>
  group_by(k, rho, n, method, rep) |>
  summarise(any = any(flag), .groups = "drop") |>
  group_by(k, rho, n, method) |>
  summarise(v = 100 * mean(any), .groups = "drop") |>
  cell_lab() |>
  rate_plot("Datasets with a fitting item flagged (%)") +
  facet_wrap(~ cell, nrow = 2)
```

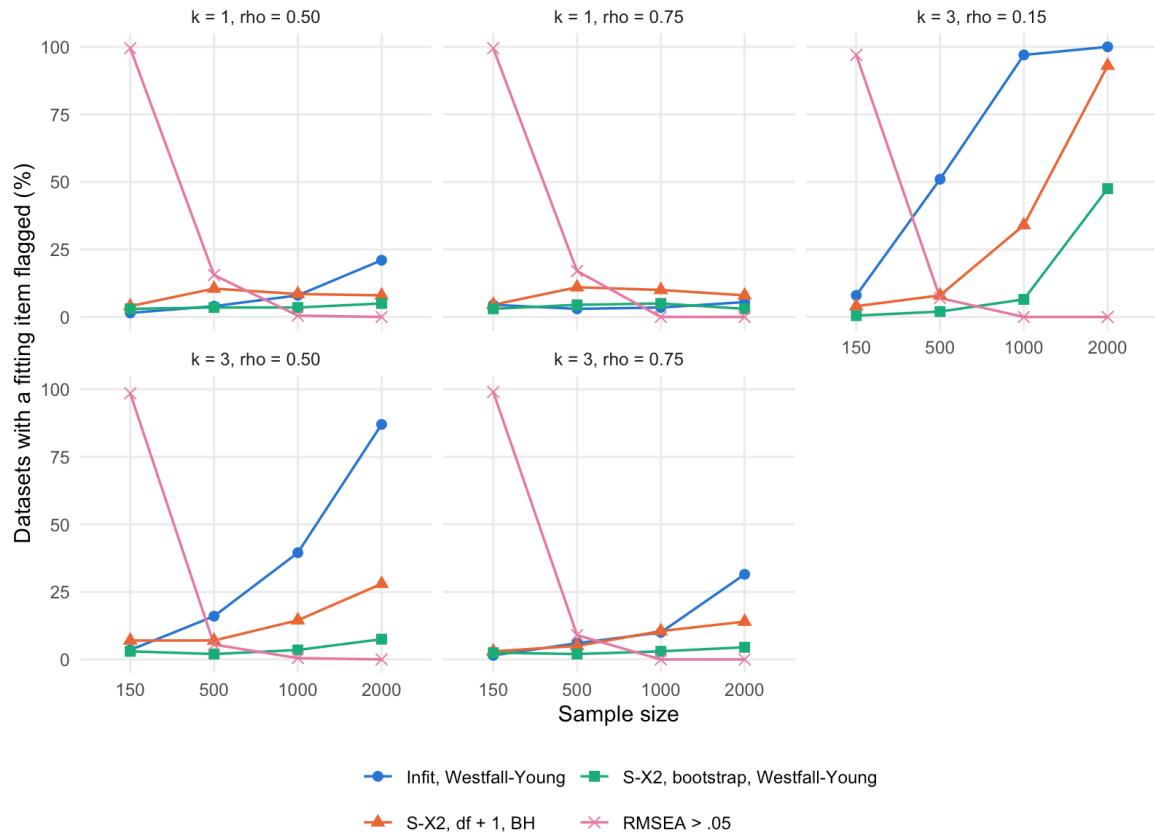


Figure 3: Share of datasets in which at least one fitting item is flagged. Panels as in Figure 2.

This is where S-X2 differs. With three misfitting items, infit flags fitting items in a share of datasets that rises steeply with n : at $\rho = 0.50$, 16 percent at $n = 500$, 39.5 at 1000 and 87 at 2000. The S-X2 bootstrap stays at 2 to 7.5 percent in the same cells. At $\rho = 0.15$ infit flags fitting items in 97 percent of datasets at $n = 1000$ against 6.5 for the S-X2 bootstrap, and only at $n = 2000$ does the S-X2 bootstrap also cascade, to 47.5 percent. With a single misfitting item the cascade is milder for infit, up to 21 percent at $\rho = 0.50$ and $n = 2000$, against 5 percent (Table 3).

The fitting items are fitting by construction, since they respond only to the first dimension. In the unidimensional model fitted to the whole set, the misfitting items distort the person locations, so fitting items can show misfit too. This is the cascade described in Johansson (2025). Flagging them is a false positive with respect to the data generating model.

Recovering the right items

```
flags |>
  filter(k > 0, method %in% fig_methods) |>
  group_by(k, rho, n, method, rep) |>
  summarise(v = all(flag[misfit]) & !any(flag[!misfit]), .groups = "drop") |>
  group_by(k, rho, n, method) |>
  summarise(v = 100 * mean(v), .groups = "drop") |>
  cell_lab() |>
  rate_plot("Perfect recovery (%)") +
  facet_wrap(~ cell, nrow = 2)
```

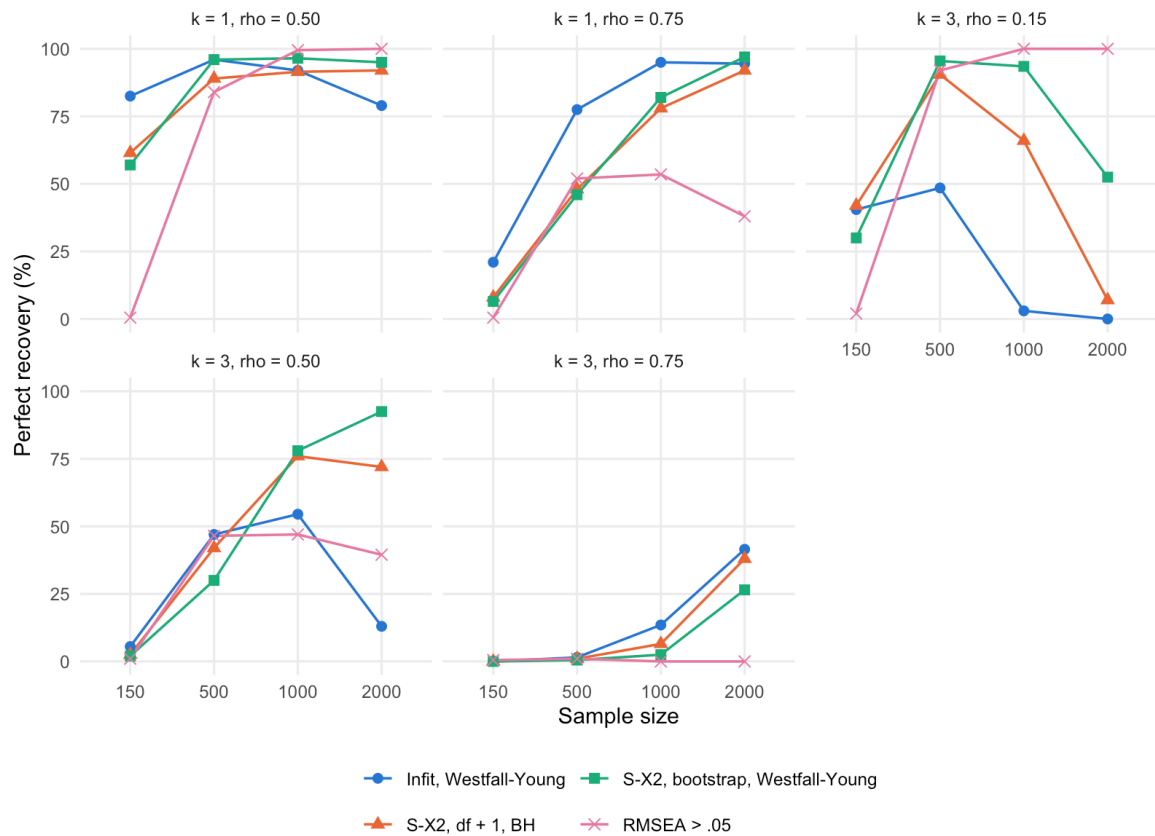


Figure 4: Share of datasets in which every misfitting item is flagged and no fitting item is. Panels as in Figure 2.

The two effects trade off. With three misfitting items the S-X2 bootstrap recovers the right set far more often at $n \geq 1000$ when misfit is strong or moderate: 93.5 against 3 percent at $\rho = 0.15$ and $n = 1000$, and 92.5 against 13 percent at $\rho = 0.50$ and $n = 2000$. Infit is better at small n and for subtle misfit, where the S-X2 bootstrap misses too many of the misfitting items: 41.5 against 26.5 percent at $\rho = 0.75$ and $n = 2000$, and 47 against 30 percent at $\rho = 0.50$ and $n = 500$ (Table 4).

Detection by item location

```
flags |>
  filter(k == 3, misfit, rho > 0.15, method %in% c("infit_wy", "sx2_boot")) |>
  mutate(location = factor(sprintf("item at %.2f logits", location),
    levels = sprintf("item at %.2f logits",
      sort(unique(location)))),
    rho = sprintf("rho = %.2f", rho)) |>
  group_by(rho, n, location, method) |>
  summarise(v = 100 * mean(flag), .groups = "drop") |>
  rate_plot("Detection (%)") +
  facet_grid(rho ~ location)
```

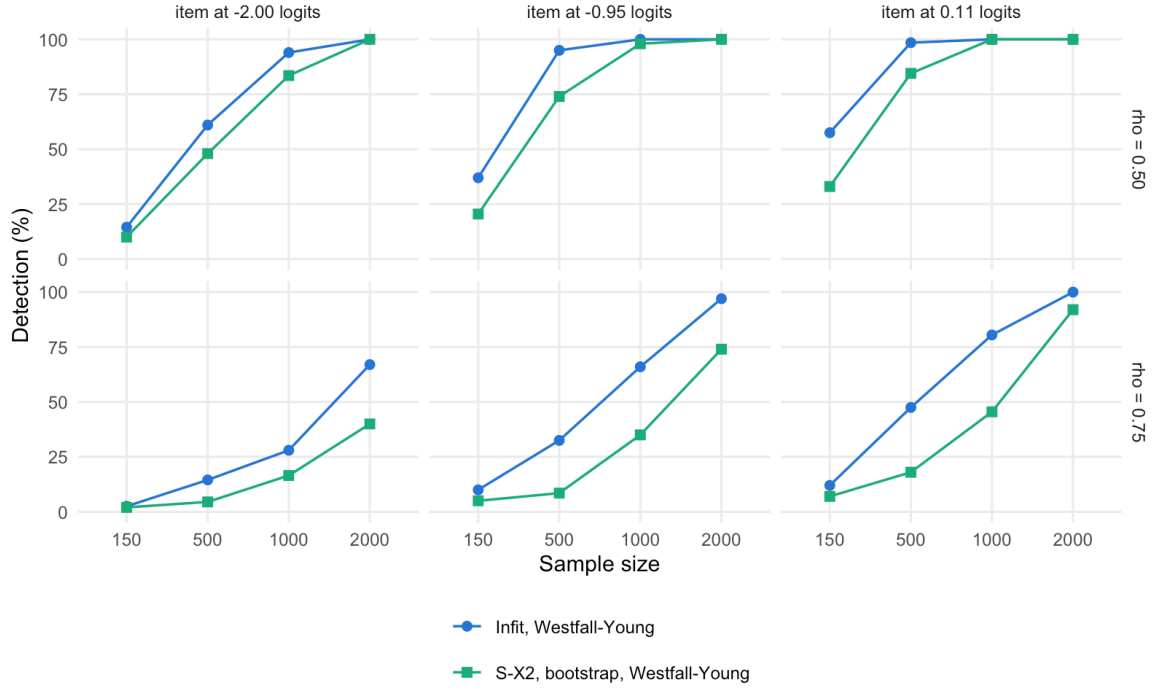


Figure 5: Detection of each of the three misfitting items by its location relative to the sample mean, for the two subtler conditions.

Both methods find it hardest to detect the item two logits below the sample mean, and the S-X2 bootstrap loses most relative to infit on the two items closer to the centre at $\rho = 0.75$ (Table 5).

At a matched null error rate

The S-X2 bootstrap is slightly above 5 percent under the null at small n and infit slightly below, which favours S-X2 in the comparisons above. In Table 6 each method's adjusted p-value threshold is set so that its family-wise rate in the null cell at the same n is 5 percent, and that threshold is applied to the misfit cells. The thresholds come from the same design, so they are an oracle.

Matching the null error rate moves detection by at most 3.3 points. False flags move more where a threshold loosens, by up to 6.5 points for infit and 13.5 for the S-X2 bootstrap, both at $\rho = 0.15$, where the cascade is steepest. Infit still detects more in every unsaturated cell, and the S-X2 bootstrap still flags far fewer fitting items when three items misfit.

Paired comparison

Both methods were run on the same datasets, so their decisions can be compared item by item. Table 7 counts misfitting items detected by one method and not the other. Where the two methods disagree about a misfitting item, it is almost always infit that detects it. The largest counts in the other direction, 15 to 17 out of 600, are in cells where infit alone detects 92 to 172.

Interpretation

1. Infit with Westfall-Young is calibrated and detects more. It holds 4.2 to 4.4 percent family-wise under the null and detects at least as much as the S-X2 bootstrap and the df-corrected S-X2 in every cell, at nominal and at matched error rates. For subtle misfit and moderate n the difference is large. The original study's conclusion 1 survives the fair comparison.

2. The S-X2 bootstrap resists the cascade. When three items misfit at $n \geq 1000$, it rarely flags fitting items, while infit flags them in most datasets. Its advantage in recovering the right set of items is large where misfit is strong or moderate and n is large. The original study's conclusion 2 survives, and the gap is wider than it reported. At $\rho = 0.15$ and $n = 2000$ infit with Westfall-Young flags 57.1 percent of fitting items, close to the 54.2 percent the original found without multiplicity correction, while the S-X2

bootstrap flags 3.7 percent against 16.9 for the original's asymptotic S-X2. Multiplicity control does not remove the cascade for infit, since the fitting items it flags there are far outside their null distribution.

3. Asymptotic S-X2 is liberal under the null, but resists the cascade. With the df corrected it is still liberal under the null, 5.8 to 9.0 percent with BH and 5.8 to 8.6 with Holm, and with mirt's df it is 8.8 to 12.4. With BH it also cascades much more than the bootstrap version, to 28 percent at $\rho = 0.50$ and $n = 2000$, but that is the correction, not the statistic. BH controls the false discovery rate and loosens its threshold once the misfitting items are rejected. Recomputed from the stored p-values with Holm, which controls the family-wise rate as Westfall-Young does, asymptotic S-X2 flags fitting items in 2.5, 4.5 and 9 percent of datasets at $\rho = 0.50$ and $n = 500, 1000$ and 2000 with three misfitting items, against 2, 3.5 and 7.5 for the bootstrap, at the same or slightly higher detection. At $\rho = 0.15$ and $n = 1000$ it is 11 against 6.5. Infit with Holm still cascades, 90.5 percent at $\rho = 0.15$ and $n = 1000$ and 72.5 at $\rho = 0.50$ and $n = 2000$. The resistance therefore belongs to the S-X2 statistic, and the bootstrap's contribution is calibration under the null. A likely reason is that the cascade leaves fitting items with small slope deviations, which infit, close to a one-df slope contrast, picks up while S-X2 spreads them over 15 to 17 df at $n = 1000$ and 2000 . The book chapter uses Holm throughout for this reason.

4. RMSEA adds nothing beyond S-X2. A bootstrap cutoff makes it usable, which confirms the original study's hypothesis, but it then behaves like the S-X2 bootstrap. A fixed threshold of .05 remains unusable.

Against the decision rule. The rule set before the run was that S-X2 is worth adding if its bootstrap version holds 5 percent under the null and, in a cell with $\rho = 0.50$ or 0.75 , detects misfit infit misses or flags clearly fewer fitting items, without a comparable loss elsewhere. The first two parts are met: the S-X2 bootstrap is at most slightly liberal at small n , and at $\rho = 0.50$ with three misfitting items it flags fitting items in 3.5 against 39.5 percent of datasets at $n = 1000$ and 7.5 against 87 at $n = 2000$, with detection 93.8 against 98 and 100 against 100. The third part is a judgement. At $\rho = 0.75$ the S-X2 bootstrap detects much less than infit at every n up to 1000.

So S-X2 is not a replacement for infit, and it is not a better default. Its use would be as a second opinion at large n : when infit flags many items, the S-X2 bootstrap separates the items that misfit from those that are flagged because others do. The cost is a parametric bootstrap with a refit per draw, about 25 to 40 seconds per analysis at $B = 400$ on one core for 20 items.

Scope and limits

- Dichotomous items only. The diagnostic found S-X2 close to nominal for polytomous items after the df correction, so a polytomous arm is the obvious next step if S-X2 goes further.
- One misfit mechanism, a second dimension. Other mechanisms, such as discrimination different from the Rasch model, may favour either method.
- S-X2 is computed by mirt under MML with a normal latent distribution, which matches the data generation here and so favours it slightly.
- The bootstrap uses mirt's df + 1 to put S-X2 on a common scale across draws. The bootstrap reference does not depend on the df otherwise.

Appendix: result tables

```
null_fw |>
  pivot_wider(names_from = n, values_from = v, names_prefix = "n = ") |>
  kb(digits = 1)
```

Table 1: Datasets with at least one flagged item under the null (%), $k = 0$.

method	n = 150	n = 500	n = 1000	n = 2000
infit_wy	4.2	4.2	4.4	4.2
infit_bh	3.8	5.0	3.4	3.8
sx2_mirt	10.0	12.4	10.2	8.8
sx2_df1	7.0	9.0	7.2	5.8
sx2_boot	6.2	6.6	4.8	3.8
rmsea_boot	6.2	6.8	4.6	3.4
rmsea_05	99.2	20.0	0.2	0.0

```

main <- c("infit_wy", "infit_bh", "sx2_df1", "sx2_boot", "rmsea_05")
flags |>
  filter(k > 0, misfit, method %in% main) |>
  group_by(k, rho, n, method) |>
  summarise(v = 100 * mean(flag), .groups = "drop") |>
  pivot_wider(names_from = method, values_from = v) |>
  kb(digits = 1)

```

Table 2: Detection, mean over the misfitting items (%).

k	rho	n	infit_wy	infit_bh	sx2_df1	sx2_boot	rmsea_05
1	0.50	150	83.5	75.5	64.5	59.0	94.0
1	0.50	500	100.0	100.0	99.5	99.5	99.5
1	0.50	1000	100.0	100.0	100.0	100.0	100.0
1	0.50	2000	100.0	100.0	100.0	100.0	100.0
1	0.75	150	22.0	18.0	9.5	7.5	66.0
1	0.75	500	80.0	74.0	55.5	47.5	62.5
1	0.75	1000	98.5	97.5	86.0	85.0	53.5
1	0.75	2000	100.0	100.0	100.0	100.0	38.0
3	0.15	150	77.8	81.7	72.3	65.0	96.7
3	0.15	500	99.7	99.8	99.5	99.2	99.5
3	0.15	1000	100.0	100.0	100.0	100.0	100.0
3	0.15	2000	100.0	100.0	100.0	100.0	100.0
3	0.50	150	36.3	37.5	25.2	21.2	73.8
3	0.50	500	84.8	90.0	77.5	68.8	80.2
3	0.50	1000	98.0	99.0	96.8	93.8	79.8
3	0.50	2000	100.0	100.0	100.0	100.0	79.2
3	0.75	150	8.2	6.3	5.0	4.7	47.5
3	0.75	500	31.5	34.3	13.3	10.3	20.0
3	0.75	1000	58.2	64.7	39.7	32.3	8.8
3	0.75	2000	88.0	93.3	78.3	68.7	1.3

```

flags |>
  filter(k > 0, !misfit, method %in% main) |>
  group_by(k, rho, n, method, rep) |>
  summarise(any = any(flag), .groups = "drop") |>
  group_by(k, rho, n, method) |>
  summarise(v = 100 * mean(any), .groups = "drop") |>
  pivot_wider(names_from = method, values_from = v) |>
  kb(digits = 1)

```

Table 3: Datasets with at least one fitting item flagged (%).

k	rho	n	infit_wy	infit_bh	sx2_df1	sx2_boot	rmsea_05
1	0.50	150	1.5	4.5	4.0	3.0	99.5
1	0.50	500	4.0	11.0	10.5	3.5	15.5
1	0.50	1000	8.0	18.5	8.5	3.5	0.5
1	0.50	2000	21.0	33.0	8.0	5.0	0.0
1	0.75	150	4.5	6.5	4.5	3.0	99.5
1	0.75	500	3.0	6.0	11.0	4.5	17.0
1	0.75	1000	3.5	7.0	10.0	5.0	0.0
1	0.75	2000	5.5	11.0	8.0	3.0	0.0
3	0.15	150	8.0	24.5	4.0	0.5	97.0
3	0.15	500	51.0	90.0	8.0	2.0	7.0
3	0.15	1000	97.0	100.0	34.0	6.5	0.0
3	0.15	2000	100.0	100.0	93.0	47.5	0.0
3	0.50	150	3.5	9.5	7.0	3.0	98.5
3	0.50	500	16.0	36.5	7.0	2.0	5.5
3	0.50	1000	39.5	73.5	14.5	3.5	0.5
3	0.50	2000	87.0	98.0	28.0	7.5	0.0
3	0.75	150	1.5	2.5	3.0	2.5	99.0
3	0.75	500	6.0	11.5	5.0	2.0	9.0
3	0.75	1000	10.0	21.0	10.5	3.0	0.0
3	0.75	2000	31.5	59.5	14.0	4.5	0.0

```

flags |>
  filter(k > 0, method %in% main) |>
  group_by(k, rho, n, method, rep) |>
  summarise(ok = all(flag[misfit]) & !any(flag[!misfit]), .groups = "drop") |>
  group_by(k, rho, n, method) |>
  summarise(v = 100 * mean(ok), .groups = "drop") |>
  pivot_wider(names_from = method, values_from = v) |>
  kb(digits = 1)

```

Table 4: Perfect recovery: every misfitting item flagged and no fitting item (%).

k	rho	n	infit_wy	infit_bh	sx2_df1	sx2_boot	rmsea_05
1	0.50	150	82.5	71.5	61.5	57.0	0.5
1	0.50	500	96.0	89.0	89.0	96.0	84.0
1	0.50	1000	92.0	81.5	91.5	96.5	99.5
1	0.50	2000	79.0	67.0	92.0	95.0	100.0
1	0.75	150	21.0	16.0	8.0	6.5	0.5
1	0.75	500	77.5	68.0	48.0	46.0	52.0
1	0.75	1000	95.0	90.5	78.0	82.0	53.5
1	0.75	2000	94.5	89.0	92.0	97.0	38.0
3	0.15	150	40.5	37.0	42.0	30.0	2.0
3	0.15	500	48.5	10.0	90.5	95.5	92.0
3	0.15	1000	3.0	0.0	66.0	93.5	100.0
3	0.15	2000	0.0	0.0	7.0	52.5	100.0
3	0.50	150	5.5	6.5	2.5	2.0	1.0
3	0.50	500	47.0	42.0	42.0	30.0	46.5
3	0.50	1000	54.5	24.0	76.0	78.0	47.0
3	0.50	2000	13.0	2.0	72.0	92.5	39.5
3	0.75	150	0.0	0.0	0.0	0.0	0.5
3	0.75	500	1.5	5.0	1.0	0.5	1.0
3	0.75	1000	13.5	18.0	6.5	2.5	0.0
3	0.75	2000	41.5	31.0	38.0	26.5	0.0

```

flags |>
  filter(k == 3, misfit, rho > 0.15, method %in% c("infit_wy", "sx2_boot")) |>
  mutate(location = round(location, 2)) |>
  group_by(rho, n, location, method) |>
  summarise(v = 100 * mean(flag), .groups = "drop") |>
  pivot_wider(names_from = method, values_from = v) |>
  kb(digits = 1)

```

Table 5: Detection by item location, $k = 3$ (%).

rho	n	location	infit_wy	sx2_boot
0.50	150	-2.0	14.5	10.0
0.50	150	-0.9	37.0	20.5
0.50	150	0.1	57.5	33.0
0.50	500	-2.0	61.0	48.0
0.50	500	-0.9	95.0	74.0
0.50	500	0.1	98.5	84.5
0.50	1000	-2.0	94.0	83.5
0.50	1000	-0.9	100.0	98.0
0.50	1000	0.1	100.0	100.0
0.50	2000	-2.0	100.0	100.0
0.50	2000	-0.9	100.0	100.0
0.50	2000	0.1	100.0	100.0
0.75	150	-2.0	2.5	2.0
0.75	150	-0.9	10.0	5.0
0.75	150	0.1	12.0	7.0
0.75	500	-2.0	14.5	4.5
0.75	500	-0.9	32.5	8.5
0.75	500	0.1	47.5	18.0
0.75	1000	-2.0	28.0	16.5
0.75	1000	-0.9	66.0	35.0
0.75	1000	0.1	80.5	45.5
0.75	2000	-2.0	67.0	40.0
0.75	2000	-0.9	97.0	74.0
0.75	2000	0.1	100.0	92.0

```

thr <- adj |>
  filter(k == 0) |>
  select(n, rep, infit_wy, sx2_df1, sx2_boot) |>
  pivot_longer(c(infit_wy, sx2_df1, sx2_boot), names_to = "method",
               values_to = "padj") |>
  group_by(method, n, rep) |>
  summarise(mn = min(padj, na.rm = TRUE), .groups = "drop") |>
  group_by(method, n) |>
  summarise(c = quantile(mn, .05, type = 1), .groups = "drop")

adj |>
  filter(k > 0) |>
  select(k, rho, n, rep, misfit, infit_wy, sx2_df1, sx2_boot) |>
  pivot_longer(c(infit_wy, sx2_df1, sx2_boot), names_to = "method",
               values_to = "padj") |>
  left_join(thr, by = c("method", "n")) |>

```

```
mutate(flag = coalesce(padj <= c, FALSE)) |>
group_by(k, rho, n, method, rep) |>
summarise(det = mean(flag[misfit]), ff = any(flag[!misfit]), .groups = "drop") |>
group_by(k, rho, n, method) |>
summarise(det = 100 * mean(det), ff = 100 * mean(ff), .groups = "drop") |>
pivot_wider(names_from = method, values_from = c(det, ff)) |>
select(k, rho, n, det_infit_wy, det_sx2_boot, ff_infit_wy, ff_sx2_boot) |>
kb(digits = 1, col.names = c("k", "rho", "n", "Detection, infit",
                             "Detection, S-X2 boot", "False flags, infit",
                             "False flags, S-X2 boot"))
```

Table 6: Detection and false flags with each method's threshold set to a 5 percent family-wise rate in the null cell at the same n (%).

k	rho	n	Detection, infit	Detection, S-X2 boot	False flags, infit	False flags, S-X2 boot
1	0.50	150	84.5	58.0	2.0	3.0
1	0.50	500	100.0	99.5	5.5	3.5
1	0.50	1000	100.0	100.0	11.0	4.0
1	0.50	2000	100.0	100.0	26.0	6.5
1	0.75	150	24.0	7.0	4.5	3.0
1	0.75	500	82.0	46.0	3.0	2.5
1	0.75	1000	98.5	85.0	5.5	5.0
1	0.75	2000	100.0	100.0	8.0	4.0
3	0.15	150	79.5	62.8	8.5	0.5
3	0.15	500	99.7	99.2	57.5	1.0
3	0.15	1000	100.0	100.0	98.5	7.0
3	0.15	2000	100.0	100.0	100.0	61.0
3	0.50	150	37.3	20.7	4.5	2.0
3	0.50	500	85.5	65.5	17.0	0.5
3	0.50	1000	98.0	94.3	44.0	3.5
3	0.50	2000	100.0	100.0	91.0	9.5
3	0.75	150	8.3	4.2	1.5	2.5
3	0.75	500	32.5	9.3	7.0	0.5
3	0.75	1000	60.8	32.7	12.0	3.0
3	0.75	2000	88.2	71.7	34.5	7.5

```
flags |>
filter(k > 0, misfit, method %in% c("infit_wy", "sx2_boot")) |>
select(k, rho, n, rep, item, method, flag) |>
pivot_wider(names_from = method, values_from = flag) |>
group_by(k, rho, n) |>
summarise(`infit only` = sum(infit_wy & !sx2_boot),
          `S-X2 boot only` = sum(!infit_wy & sx2_boot),
```

```
both = sum(infit_wy & sx2_boot), .groups = "drop") |>
  kb()
```

Table 7: Misfitting items detected by one method and not the other, infit with Westfall-Young against the S-X2 bootstrap, same datasets.

k	rho	n	infit only	S-X2 boot only	both
1	0.50	150	51	2	116
1	0.50	500	1	0	199
1	0.50	1000	0	0	200
1	0.50	2000	0	0	200
1	0.75	150	31	2	13
1	0.75	500	65	0	95
1	0.75	1000	28	1	169
1	0.75	2000	0	0	200
3	0.15	150	92	15	375
3	0.15	500	4	1	594
3	0.15	1000	0	0	600
3	0.15	2000	0	0	600
3	0.50	150	107	16	111
3	0.50	500	108	12	401
3	0.50	1000	27	2	561
3	0.50	2000	0	0	600
3	0.75	150	31	10	18
3	0.75	500	133	6	56
3	0.75	1000	172	17	177
3	0.75	2000	125	9	403

Reference

Johansson, M. (2025). Detecting item misfit in Rasch models. *Educational Methods & Psychometrics*, 3(18). <https://doi.org/10.61186/emp.2025.5>